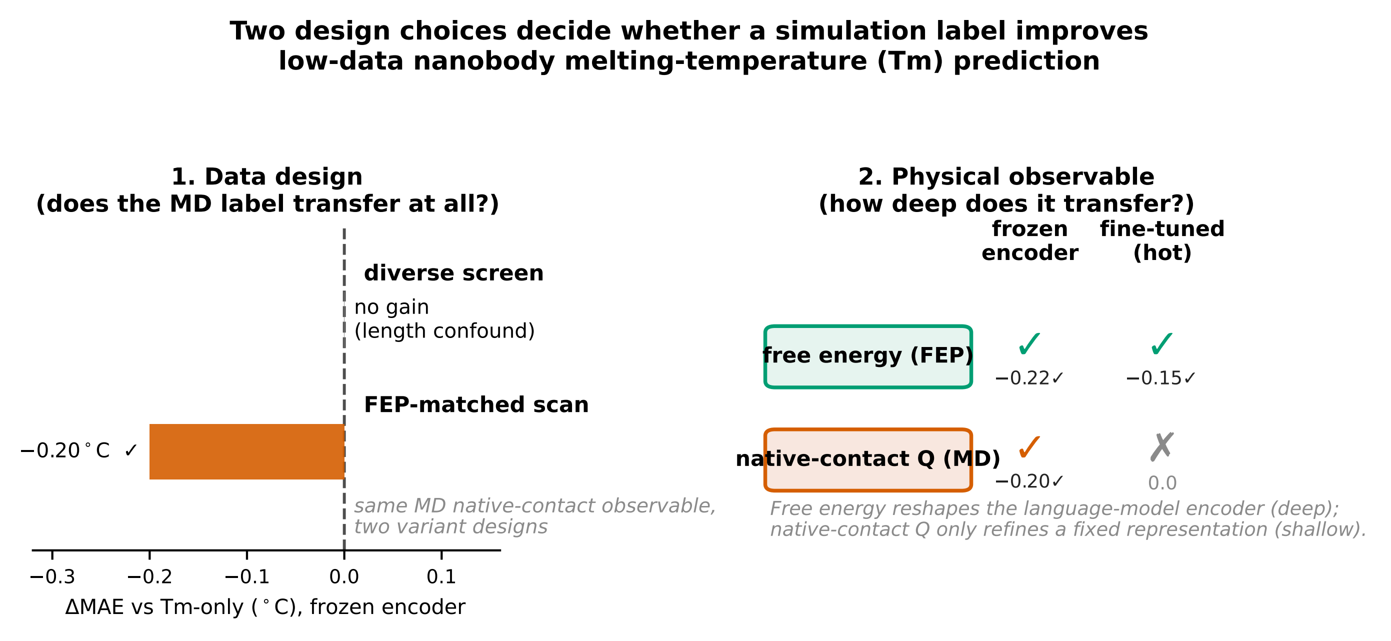
Two design choices decide whether a simulation label improves
low-data nanobody melting-temperature (Tm) prediction
1. Data design
(does the MD label transfer at all?)
diverse screen
no gain
(length confound)
FEP-matched scan
−0.20 °C ✓
same MD native-contact observable,
two variant designs
−0.3
−0.2
−0.1
0.0
0.1
ΔMAE vs Tm-only (°C), frozen encoder
2. Physical observable
(how deep does it transfer?)
frozen
encoder
fine-tuned
(hot)
free energy (FEP)
✓
−0.22✓
✓
−0.15✓
native-contact Q (MD)
✓
−0.20✓
✗
0.0
Free energy reshapes the language-model encoder (deep);
native-contact Q only refines a fixed representation (shallow).